The tectonic relationship of two Early Proterozoic dyke swarms to the Kapuskasing Structural Zone: a paleomagnetic and petrographic study[1]

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A tea-coloured cloudiness in groundmass feldspars characterizes dykes of two Early Proterozoic swarms (2.45 Ga Matachewan; 2.04 Ga Kapuskasing) that intrude amphibolite to granulite host rocks within the Kapuskasing Structural Zone (KSZ) of northern Ontario. Outside the KSZ the dykes occur in lower-grade terranes and contain unclouded feldspars. Matachewan dykes within the KSZ are also fresh, contain an aluminous, blue–green amphibole, have fewer plagioclase phenocrysts, and on average are thinner compared with their counterparts outside, which are more hydrously altered, contain a green, fibrous amphibole, and are locally abundant in felspar phenocrysts.

Within the Matachewan swarm, magnetic polarity domains — areas of dominantly normal or reversed magnetization — have been identified; the boundaries of these domains correspond closely to major faults. Two of these faults form the boundaries of the KSZ and separate normal dykes within from dominantly reversed ones outside.

The foregoing petrological and paleomagnetic observations are related to differential vertical crustal movements that have raised younger, normal-polarity Matachewan dykes (as exemplified by those within the KSZ) from a deeper crustal level and juxtaposed them against older, reversely magnetized dykes that were intruded at shallower crustal levels.

Paleomagnetic baked-contact tests on Kapuskasing dykes suggest that they were intruded both before and after the main period of crustal uplift and tilting along the KSZ and that dykes within 4 km of the southeastern boundary fault were raised from depths where temperatures exceeded 580°C, the Curie-point isotherm for magnetite.

Un aspect trouble, de coleur thé, dans les feldspaths de la matrice caractérise les dykes de deux essaims d'âge protérozoïque (Matachewan, 2,45 Ga; Kapuskasing, 2,04 Ga), lesquels recoupent les roches de faciès des amphibolites à granulites, dans la zone structurale de Kapuskasing (ZSK) du nord de l'Ontario. En dehors de ZSK, les dykes traversent des terrains moins métamorphisés et renferment des feldspaths clairs. Les dykes de Matachewan de l'intérieur de la ZSK sont également frais, renferment une amphibole bleu–vert, les phénocristaux de plagioclase y sont plus rares, et en moyenne ils sont plus étroits que les dykes constituant la contrepartie extérieure, ces derniers sont plus altérés par l'eau, contiennent une amphibole fibreuse et exhibent localement une forte abondance de phénocristaux de plagioclase.

Dans l'essaim de Matachewan, des domaines de polarité magnétique — régions d'aimantation normale ou inversée – ont été identifiés, et leurs frontières coïncident d'assez près avec les failles majeures. Deux de ces failles délimitent la ZSK et séparent les dykes de polarité normale de l'intérieur de la ZSK d'avec ceux de polarité, principalement inversée, situés à l'extérieur.

Les observations pétrologiques et paléomagnétiques précédentes sont associées aux mouvements crustaux verticaux différentiels qui ont soulevèrent les dykes de Matachewan plus jeunes de polarité normale, par exemple ceux de l'intérieur de la ZSK, en partant d'un niveau plus profond et jusqu'à leur juxtaposition avec les dykes plus âgés de polarité inversée, dont la mise en place a eu lieu à des niveaux crustaux moins profonds.

Les tests du contact appliqués sur les dykes de Kapuskasing suggèrent leur mise en place tant avant qu'après la période principale de soulèvement et basculement crustal le long de la ZSK et que les dykes situés à l'intérieur d'une bande de 4 km de la frontière sud-est de la faille furent soulevés à partir de profondeurs où la température était plus élevée que 580°C, le point de Curie de la magnétite.

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Introduction

The Kapuskasing Structural Zone (KSZ) in northern Ontario (Fig. 1) is a north-northeast-trending, linear, fault-bounded region that contains high-grade metamorphic rocks and transects the dominantly east–west structural grain of the Superior Province (Percival and Card 1983). The eastern margin of the zone has been interpreted by these authors as a west-dipping reverse-fault zone (the Ivanhoe Lake Cataclastic Zone, ILCZ) along which Archean crust has been upthrust to the east. Granulite-facies metamorphic rocks found discontinuously along the KSZ are thus thought to represent uplifted crust. From Wawa in the west to Chapleau at the southern end of the KSZ an approximately 20 km thickness of crust is exposed in an oblique, westerly dipping section (Percival and Card 1985). From the deepest, granulitic rocks exposed in the KSZ the crust successively passes up into the Wawa domal gneiss terrane and the Michipicoten granite–greenstone belt as metamorphic grades generally decrease through amphibolite to greenschist facies.

Two sets of Precambrian mafic dykes are present in the vicinity of the KSZ: the north- to northwest-trending Matachewan–Hearst dykes (Ernst and Halls 1984; age 2454 ± 2 Ma, Heaman 1988) and the east-northeast-trending Kapuskasing dykes (Card 1981; age 2043 ± 14 Ma, Hanes *et al.* 1988).

The Matachewan–Hearst swarm has a total length of about 700 km and fans northwards from a focal region along the southern margin of the Superior Province (Fig. 1). It includes

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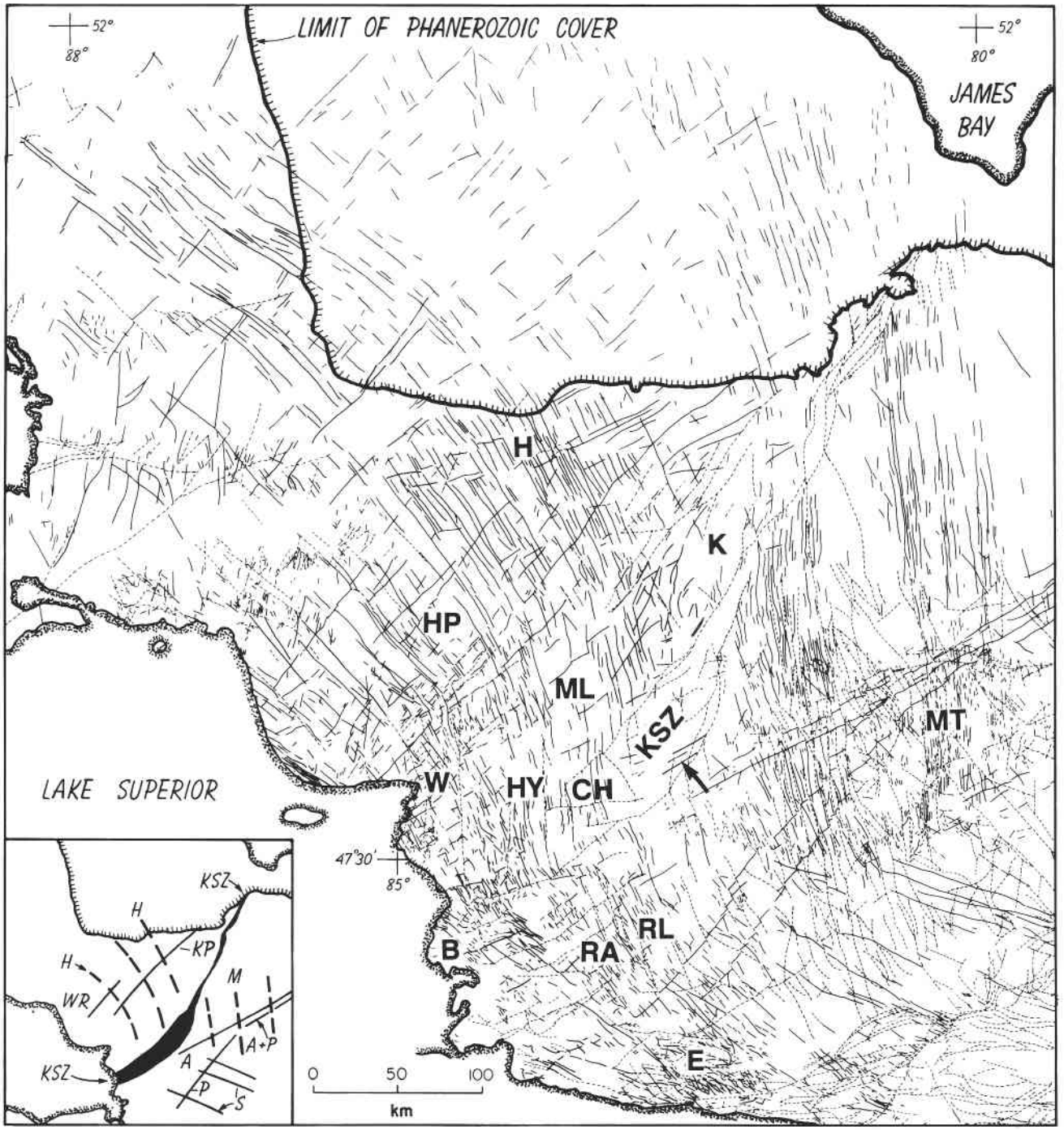


FIG. 1. Map of Precambrian dyke swarms of northern Ontario (modified after Ernst and Halls 1984) showing the location of areas sampled for Matachewan dykes. The solid lines denote dykes; the dashed lines, faults. HP, Highway 631 through Hornepayne; HY, Highways 101 and 651 west of the Kapuskasing Structural Zone (KSZ); H, Hearst; CH, Chapleau; ML, Missinaibi Lake; RL, Rocky Island Lake; RA, Ranger Lake; B, Batchawana. Results from HY, CH, and ML dykes are reported in this paper, although petrographic and polarity information is used from the other studies (B and RA from Halls and Shaw 1988; RL from H. C. Halls and M. P. Bates, unpublished data; HP and H from H. C. Halls, unpublished data, and Ernst and Halls 1984). The arrow locates the Kapuskasing dykes. Other locations: E, Elliot Lake (near Rawhide Lake); K, Kapuskasing; MT, Matachewan; W, Wawa. Major dyke swarms shown in inset: M, Matachewan (2.45 Ga); H, Hearst (2.45 Ga); P, Preissac (2.14 Ga); WR, White River (2.14 Ga); S, Sudbury (1.24 Ga); A, Abitibi (1.14 Ga); KP, Kipling (1.14 Ga).

three subswarms that are clearly delineated from aeromagnetic data; from east to west the subswarms show increasing deflection as they cross the KSZ (Fig. 1). These changes in dyke trend suggest that the tectonic development of the KSZ has influenced the dyke pattern. Two explanations are that the dykes were rotated around more or less horizontal axes during westward tilting of the crust as it was thrust upwards to the east along the ILCZ (Percival and Card 1983; Ernst and Halls 1984) or were rotated in a clockwise sense around vertical axes in response to dextral shear along the KSZ. Either explanation or a combination of both is feasible, at least chronologically, because radiometric evidence suggests that the main period of

uplift along the KSZ was between 1950 and 1900 Ma (Percival 1988), at least 400 Ma after emplacement of the Matachewan–Hearst (henceforth "Matachewan") dykes. A third explanation, proposed by Watson (1980), is that the dykes were intruded along extensional zones associated with sinistral shear along the KSZ but were not deformed or rotated after emplacement.

The Kapuskasing dykes (arrowed in Fig. 1) form a small and relatively ill-defined swarm that has achieved prominence in the context of KSZ studies because it occurs within the granulite-grade rocks at the southern end of the KSZ and therefore may be particularly useful in estimates of crustal uplift and tilting. All east-northeast-trending dykes that occur in the vicinity of the KSZ are included as part of the Kapuskasing swarm, but as discussed later, more than one age of intrusion may be represented.

Paleomagnetic studies (e.g., Irving and Naldrett 1977; Ernst and Halls 1984) show that Matachewan dykes yield a pole position that plots in a unique region of the North American apparent polar wander path and possess either normal or reversed magnetic polarity. During this study we have discovered segments of the swarm, several tens of kilometres in lateral dimension, in which the dyke population is dominantly of either polarity.

Our principal objective is to document the distribution of these magnetic-polarity domains within the Matachewan swarm and to interpret their origin in terms of differential crustal uplift and the age and depth of emplacement of the dykes. A secondary objective is to test the extent to which the Matachewan and Kapuskasing dykes have been involved in KSZ tectonics. Our study is based on paleomagnetic data from more than 600 oriented samples collected at 68 sites from the western half of the Matachewan swarm and from the Kapuskasing dykes in the vicinity of Chapleau, along Highway 101 to the west and northwards as far as Missinaibi Lake (Fig. 2).

Field and analytical procedure

Previous experience with Early Proterozoic dykes (Halls 1986) showed that fresh chilled margins retain a primary remanence more often than coarser grained interiors. Accordingly, samples were obtained from chilled margins wherever possible, with a minority taken from dyke interiors. Typically six to eight oriented samples were collected from each site with both magnetic and astro compasses. Samples were mostly field drilled along Highway 101 and block sampled elsewhere. At some sites where cross-cutting dykes were observed, as many as 20 oriented samples were obtained. The attitude and thickness of each sampled dyke were recorded where possible, together with any information relating to alteration, secondary veining, and dyke structure. The distribution of sites (Fig. 2) is shown in more detail in Fig. 3 for a 70 km stretch of highway west of Chapleau where virtually every exposed dyke has been mapped and the majority sampled.

The natural remanent magnetization (NRM) of specimens prepared from field samples was measured on spinner magnetometers (a modified DIGICO at the University of Toronto and a Schonstedt DSM-1 at The University of Western Ontario) and stepwise demagnetized in an alternating field (AF) having a peak intensity of 100 mT. A limited amount of thermal demagnetization was performed using a Schonstedt TSD-1 furnace. Remanence directions were obtained individually for each specimen following a procedure (Halls 1986) that uses remagnetization circles (Bailey and Halls 1984) and principal-component analysis (Kirschvink 1980; mean angular deviation values $\leq 10°$). A summary of all structural and paleomagnetic data obtained is given in the Appendix.

High-field Curie points for which the saturation magnetization was measured as a function of temperature were determined for several Matachewan dykes with a horizontal pendulum balance of the type depicted by Collinson (1983, Fig. 3.11).

Petrology

Matachewan dykes

Matachewan dykes are iron-rich tholeiites characterized by phenocrysts of Ca-rich plagioclase (An_{80-85}) that may be up to 6 cm across and that were present in the magma at the time of dyke injection (Phinney *et al.* 1988). Their distribution in the dykes is variable: they may be scattered evenly throughout the dyke or concentrated in one or more discrete, dyke-parallel zones that may occur anywhere from the margin to the centre of the dyke.

At the microscopic level, groundmass plagioclase and clinopyroxene occur in subequal amounts and most commonly display an intergranular texture (Figs. 4A–4C). Additional primary minerals, in accessory amounts, are Fe–Ti oxides, biotite, and apatite. In some samples, small amounts of interstitial quartz form a myrmekitic texture with feldspar (Fig. 4B). In dykes within middle to upper amphibolite-grade country rocks, such as in the Chapleau, Hornepayne, and Lake Missinaibi areas, a deep green to blue–green amphibole rims pyroxene and also occurs as small grains between plagioclase laths (Fig. 4C). The amphibole thus shows features suggestive of both secondary growth and primary magmatic crystallization. High Al_2O_3 values (9–10% at site 170 and about 7% at site 195) have been recorded from the amphiboles in the vicinity of the KSZ (Palmer and Barnett 1988). South of the KSZ where host-rock grades are greenschist to amphibolite, amphiboles are green and fibrous and have lower Al_2O_3 contents of about 4–5% (H.C. Palmer and R. L. Barnett, unpublished data). Although Matachewan dykes are generally fresh in the Chapleau region, variable amounts of alteration expressed in the production of sericite, chlorite, amphibole, epidote, prehnite, sphene, and calcite are typical of dykes in other areas (Fig. 4D). To the west along Highway 101 and to the north along Lake Missinaibi, dykes are altered locally, sometimes intensely so, but most have only a minor development of secondary minerals and are thus similar to the Chapleau dykes. In the Hornepayne area the dykes along a 150 km traverse from White River to Hearst have a moderately fresh appearance, with more frequent patches of hydrous alteration than the dykes in the Chaplea area. South of the KSZ the alteration state changes dramatically. Plagioclase twinning is still discernible, but laths are heavily altered by secondary sericite, chlorite, sphene, and other minerals. There is also a marked increase in chlorite and the fibrous, low-Al amphibole replacing clinopyroxene. These high alteration levels are found in the Batchawana, Ranger Lake, Rocky Island Lake, Rawhide Lake, and Matachewan areas (see Fig. 1 for locations). The occurrence of chlorite and a low-Al or actinolitic amphibole imparts a pronounced greenish colour to the dykes in these regions.

Kapuskasing dykes

Kapuskasing dykes are seldom porphyritic megascopically. A minority are microporphyritic in plagioclase and very sparse olivine (Fig. 4E). In thin section nonporphyritic dykes comprise subequal amounts of plagioclase and clinopyroxene, with the latter mineral rimmed by a deep-green hornblende; accessory amounts of Fe–Ti oxides and biotite are present. Where olivine

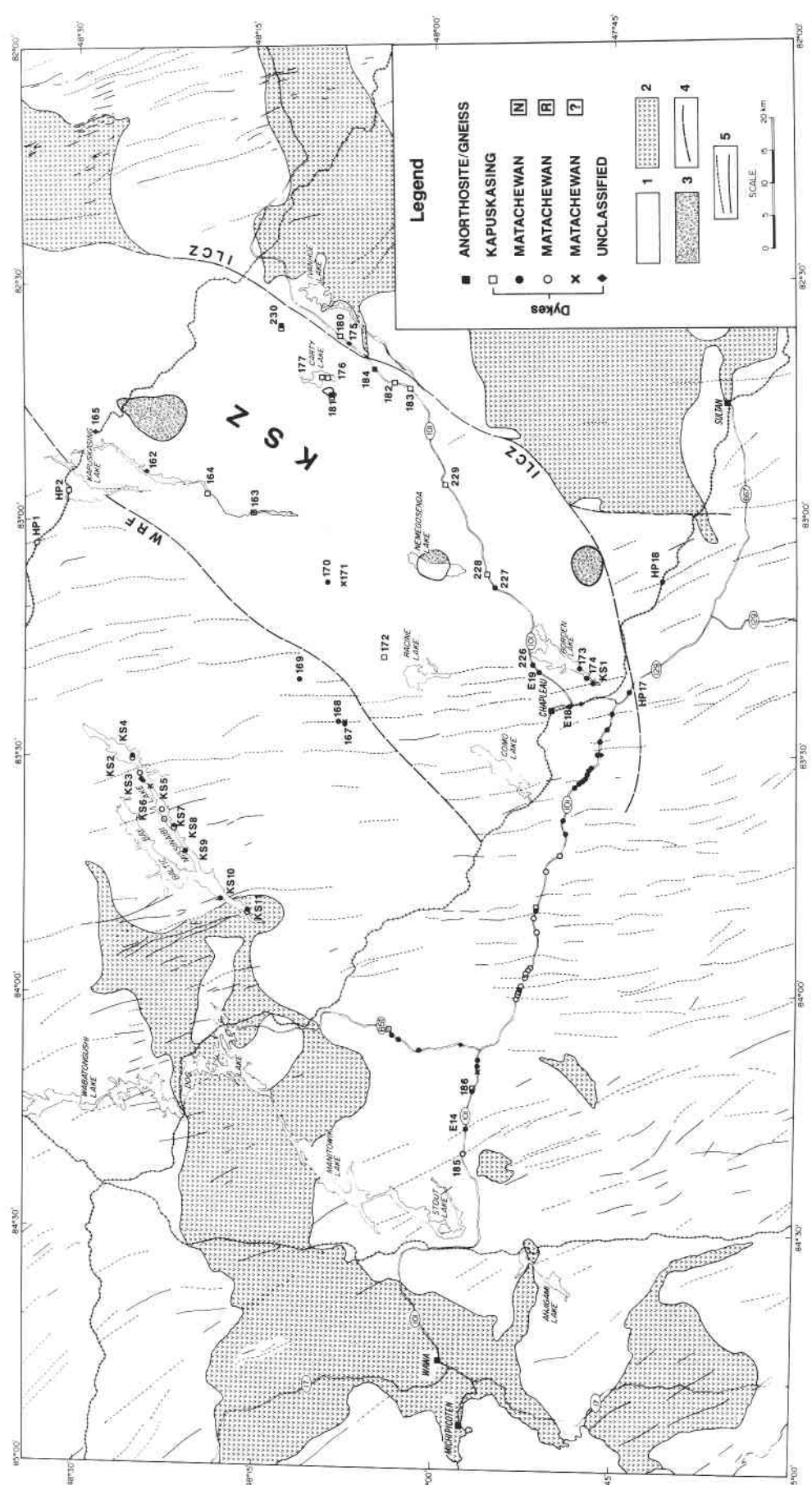


FIG. 2. Geological map, based on Ontario Geological Survey maps 2220 and 2221 (OGS 1972, 1976), showing the location of all sampling sites in the vicinity of the KSZ. Sites in the densely sampled region along Highway 101 west of Chapleau and along Highway 651 are shown in Fig. 3. Matachewan dykes are defined as those that trend north-northwest and contain green feldspar phenocrysts and (or) a typical Matachewan paleomagnetic direction. The dykes are subdivided on whether they have normal (N), reversed (R), or unknown (?) magnetization. Kapuskasing dykes are those with a general east-northeast trend. Sites where dykes cross-cut one another are shown as compound symbols. Numbers in legend: 1, Archean granite and gneiss; 2, Archean greenstone and metasedimentary rocks; 3, alkaline intrusions; 4, major faults (the Ivanhoe Lake Cataclastic Zone (ILCZ) defines the eastern boundary of the KSZ; the Wakusimi River fault (WRF) defines the western, although it is poorly defined towards the southwest); 5, Matachewan dykes, inferred from aeromagnetic data and defined from geological mapping.

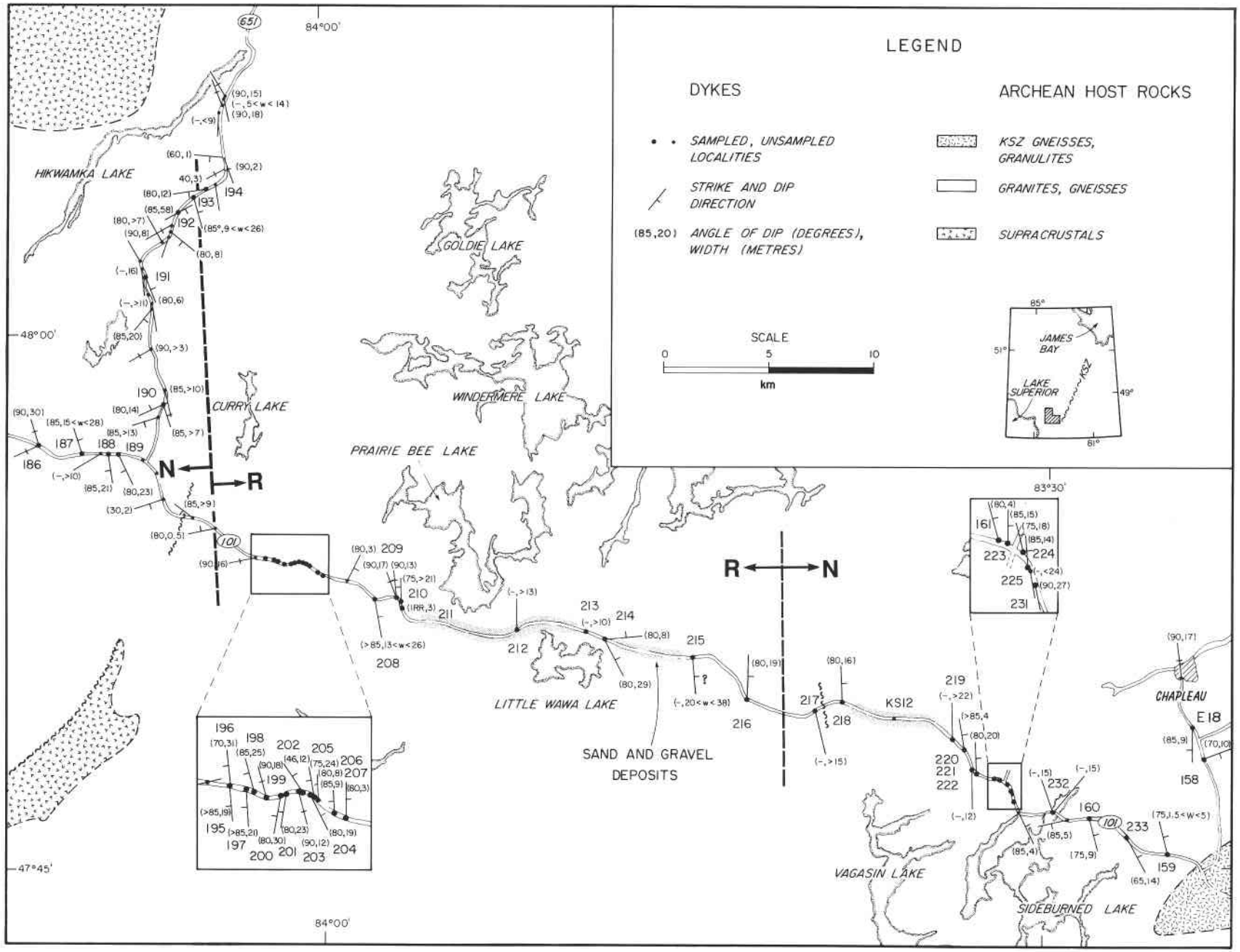


FIG. 3. Map of all exposed dykes along a 70 km traverse along Highways 101 and 651 showing the location of sampling sites. Major geological units after Ontario Geological Survey maps 2220 and 2221 (OGS 1972, 1976). All sites except 186, 194, and 214 are exclusively within Matachewan dykes. Dashed lines mark boundaries between zones in which Matachewan dykes are dominantly of one polarity, either reversed (R) or normal (N). These boundaries occur within a few kilometres of faults (the eastern one from Ontario Geological Survey Map 2221; the western one from Moser 1988).

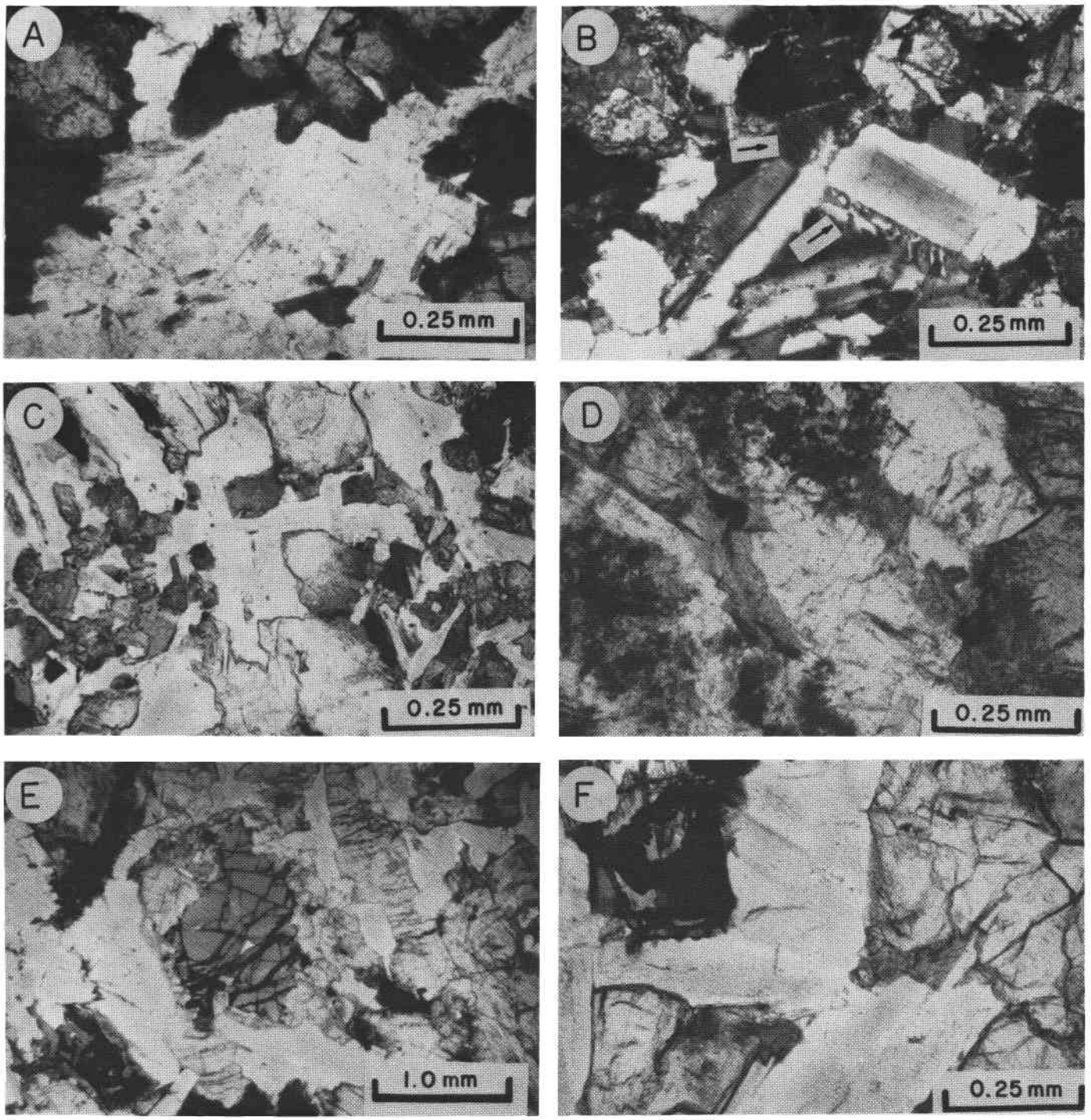


FIG. 4. Photomicrographs of Matachewan (A, B, C, D, and F) and Kapuskasing (E) dykes. B is with crossed polars; all others are in plane-polarized light. (A, B) Sample 232-3, showing oxides (black), clinopyroxene with overgrowths of amphibole, and plagioclase with slight clouding; the arrows in B indicate interstitial areas of myrmekite. (C) Sample 170-4, showing cloudy plagioclase, overgrowths of amphibole on clinopyroxene, and separate development of aluminous amphibole (grey colour). (D) Sample 216-4, showing sphene–sericite–clinozoisite alteration of plagioclase. (E) Sample 181-8, showing olivine phenocryst mantled by pyroxene and plagioclases with clouded cores and clear rims. (F) Sample 173-2, showing Fe–Ti oxides mantled by biotite; intensely clouded plagioclase and clinopyroxenes with marginal overgrowths of hornblende.

is lacking, minor interstitial quartz is sometimes present, and the dykes are indistinguishable from Matachewan dykes at their eastern limit in the Chapleau area. The presence of minor olivine in some dykes (sites 172, 181, and 229) and minor quartz in others (sites 163, 176, and 230) suggests that two distinct magma types are involved.

A curious feature of the Kapuskasing dykes where they are emplaced in amphibolite- to granulite-grade host rocks (sites 163, 164, 176, 181, 183, 229, and 230) is that their plagioclase shows an unusual tea-colour cloudiness under plane-polarized light (Figs. 4E, 4F). This phenomenon is also observed in a northeast-trending dyke at site 162, also in high-grade rocks. However, immediately outside the KSZ, where host-rock metamorphic grades suddenly drop, Kapuskasing dykes show

unclouded feldspars (e.g., site 180, 1 km southeast of the ILCZ and site HP2, 3 km northwest of the Wakusimi River fault, Fig. 2). Towards the west, where host metamorphic grades decline more gradually, the cloudiness also disappears (e.g., at sites 172, 214, and 186). A similar pattern also holds for the Matachewan dykes. Along Highway 101 they have unclouded feldspars, but approximately eastwards from site 232, where the metamorphic grade in the host rocks begins to increase within the KSZ, the cloudiness level increases to a maximum at sites 170, 171, 173, 174, 226, and KS1 (Figs. 2, 3 and cf. Figs. 4A, 4C, and 4F).

Clouded plagioclases are the result of minute inclusions of spinel, magnetite, or other minerals that develop in the plagioclase after crystallization. The conditions that produce the clouding are not well understood (Poldervaart and Gilkey 1954), but our observations show that it occurs in diabase dykes only where their enclosing host rocks are at upper amphibolite to granulite grades of metamorphism. Dykes preserving their original form and texture and containing cloudy feldspars have been reported before in host rocks of high metamorphic grade from India (Pichamuthu 1959), Fennoscandia (Solyom 1988), and Scotland (Bailey 1951). We propose therefore that cloudy feldspars in fresh, undeformed dykes originally crystallized and perhaps were subsequently altered at deep crustal levels where middle to upper amphibolite grades of metamorphism prevailed.

Paleomagnetic results

Matachewan dykes

The Matachewan dykes contain a well-defined remanence component that may have either of two approximately antipodal directions (Strangway 1961, 1964; Fahrig *et al.* 1965). One direction, designated as normal polarity (N), has a shallow positive inclination and north-northeast declination, and the opposite one is referred to as having reversed polarity (R). On the basis of partial baked-contact tests on single dykes of each polarity in the Timmins–Matachewan area (R, Pesonen 1973; N, Irving and Naldrett 1977), both components appear to be of primary origin, i.e., thermal and (or) deuteric chemical remanences formed during the initial cooling of the dyke.

Typical responses to AF demagnetization of N and R dykes (Fig. 5) show that the component is often isolated after intensities have fallen to below 10% of the initial NRM value. In many sites, chilled margins have higher median destructive fields than coarser grained interior samples and more frequently yield well-defined characteristic component linear segments on vector diagrams. The R component is often defined with less precision than the N component, both at the sample and within-site levels. Where N and R dykes are intermixed, the average α_{95} values for 18 R and 16 N sites are, respectively, 12.4° ± 5.8° (1σ) and 7.6° ± 2.7°. This difference may be a reflection of a higher level of alteration of R dykes than of N dykes and may indicate the older age of the R dykes. Alternatively, the imperfect removal of overprints, most of which have steep positive inclinations, would have a larger scattering effect on the R component.

Thermal demagnetization was carried out on four specimens showing clear N and R components, and the results indicate that the remanence is unblocked at about 585°C for both polarity states (Fig. 6). Thermomagnetic curves (lower diagrams in Fig. 6) show Curie points at about 585°C and reversible curves on heating and cooling for samples carrying either the N or R components. Both thermal-demagnetization and Curie-point determinations therefore reveal no evidence of any magnetizable phase other than titanium-poor magnetite. The similarity in behaviour of the thermal data suggests that the N and R samples have had comparable histories of remanence acquisition and alteration and, in harmony with the baked-contact tests, suggests that both components, at least in the KSZ region, were formed during the original cooling of the dykes.

In the Chapleau area, approximately at the western end of the KSZ, a segment of the Matachewan swarm, more than 50 km wide and containing at least 20 dykes, is entirely of N polarity (Fig. 2). This N-polarity zone is followed to the west by another about 25 km wide of predominantly R polarity, and this in turn is followed by a 25 km wide zone of almost exclusively N-polarity dykes (Fig. 3). This pattern could reflect the lateral growth of a dyke swarm over a time span that includes two field reversals, but cross-cutting relations between Matachewan dykes of opposite polarity (sites KS4 and KS11 along Lake Missinaibi) are consistent with a single R → N reversal during swarm emplacement (Halls and Palmer 1988; Halls 1989). Also, the boundaries between the R and N zones appear to closely coincide with faults (Fig. 3), suggesting that the polarity zonation may have a tectonic cause.

The abundance of N dykes along Highway 101, particularly in the Chapleau area, is in sharp contrast with that found in other sampling traverses to the north and south of the KSZ. In the Hornepayne (H.C. Halls, unpublished data), Batchawana, and Ranger Lake areas (Halls and Shaw 1988) and along Rocky Island Lake (Bates and Halls 1988), R dykes outnumber N dykes by at least four to one. Along Missinaibi Lake there are equal numbers of R and N dykes. These dykes are directly along strike to the north from those in the Chapleau area, although no individual dyke correlations can yet be made. In terms of the total thickness of dyke injection, the R episode exceeds the N episode along Lake Missinaibi (Appendix), and yet only 50 km to the south, in the Chapleau area, dykes are exclusively N. Near Rawhide Lake, north of Elliot Lake, Matachewan dykes emerge from beneath the Huronian sequence and are dominantly of R polarity (Aibangbee 1982). These dykes and those along Rocky Island Lake appear from Fig. 1 to be approximately on strike with those in the Chapleau area. In the Matachewan–Timmins area, within the central and eastern subswarms, 11 of 12 dykes sampled by Irving and Naldrett (1977) and Pesonen (1973) and ten dykes sampled by H.C. Palmer (unpublished data) have R polarity.

For 57 dykes studied paleomagnetically from the Chapleau and Lake Missinaibi areas, the ratio of nonporphyritic to porphyritic types is 23:13 for N dykes and 8:13 for R dykes (see Appendix). Similar ratios were found for dykes along Rocky Island Lake (Bates and Halls 1988). If only one reversal of the Earth's magnetic field occurred during emplacement of the Matachewan swarm, these data suggest a reduction in the relative frequency of porphyritic versus nonporphyritic magma injection as the swarm evolved. A possible explanation for this change is that parent magma chambers became deeper than about 10 km, the crustal level below which plagioclase megacrysts of appropriate compositions are not produced in Fe-rich tholeiites similar to Matachewan melts (Phinney and Morrison 1988).

A summary of paleomagnetic results from the KSZ area (Fig. 7; Table 1) shows that in any one region dyke directions are closely grouped but that small differences, significant at the 95% confidence level, exist between the R site-mean values for different regions (compare, for example, HY and RA in Fig. 7).

Kapuskasing dykes

The distribution of site paleomagnetic directions for east-

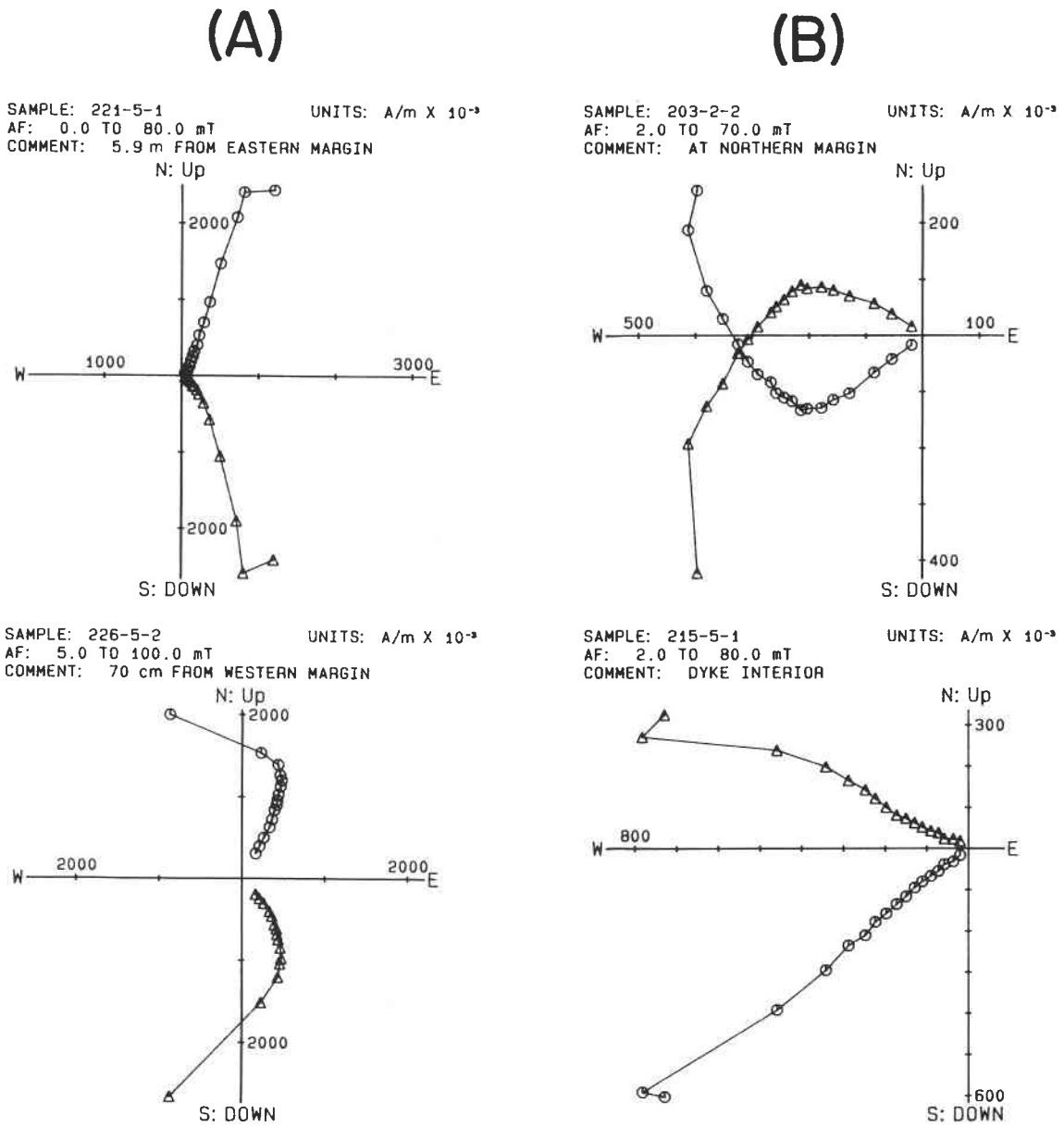


FIG. 5. Orthogonal vector plots in diagrams (A) and (B) show, respectively, AF demagnetization behaviour of N and R Matachewan dykes. Circles represent the position of the end of the magnetization vector projected onto a horizontal plane; triangles, the projection onto an east–west vertical plane. Linear segments occurring simultaneously on both plots indicate the removal of a single component; if they decay to the origin, one component remains and a stable end point in direction is achieved.

northeast-trending Kapuskasing dykes (Fig. 8) shows a clustering of directions with steep negative inclination and southeast declination but with a pronounced streaking in a northwest direction. The vertical dyke at site 180 on the footwall side of the ILCZ yields a direction within the cluster. However, the remanence from dykes found in granulite-grade rocks within 4 km of the ILCZ (sites 182, 183, and 230) plot within the northwest quadrant of the stereonet. At all three sites the dykes dip about 70°SE, and thus the paleomagnetic and structural data suggest that they were once vertical and have become inclined to the southeast in response to a 30°NW crustal tilting adjacent to the ILCZ. Granulite host rocks at site 184 containing pseudotachylyte veins and within 1 km of the ILCZ also yield a remanence direction with northwest declination. At site 182, a baked-contact test is negative: the granulite host rocks retain the same direction as the dyke out to distances of six times the dyke width (Vandall 1986). At site 230, anorthosite samples at distances of up to 1.5 times the dyke width have virtually the same direction as the dyke (Fig. 9), implying in common with site 182 that both were uplifted, acquired a remanence on passing through their unblocking temperatures, and then were tilted during upthrusting adjacent to the ILCZ. Thermal demagnetization of the anorthosites (Constanzo-Alvarez 1986) and Kapuskasing dykes (H.C. Palmer, unpublished data) suggests that single-domain magnetite is the dominant remanence carrier, so both dyke and host rocks acquired their remanences at similar temperatures and times. At distances greater than about 4 km north of the ILCZ, there is no evidence that any east-northeast-trending dykes have been rotated, if site 180 is used as the undisturbed reference direction. At site 181, a vertical, 13 m wide dyke has a steep upward remanence direction to the southeast, similar to that at site 180, but the

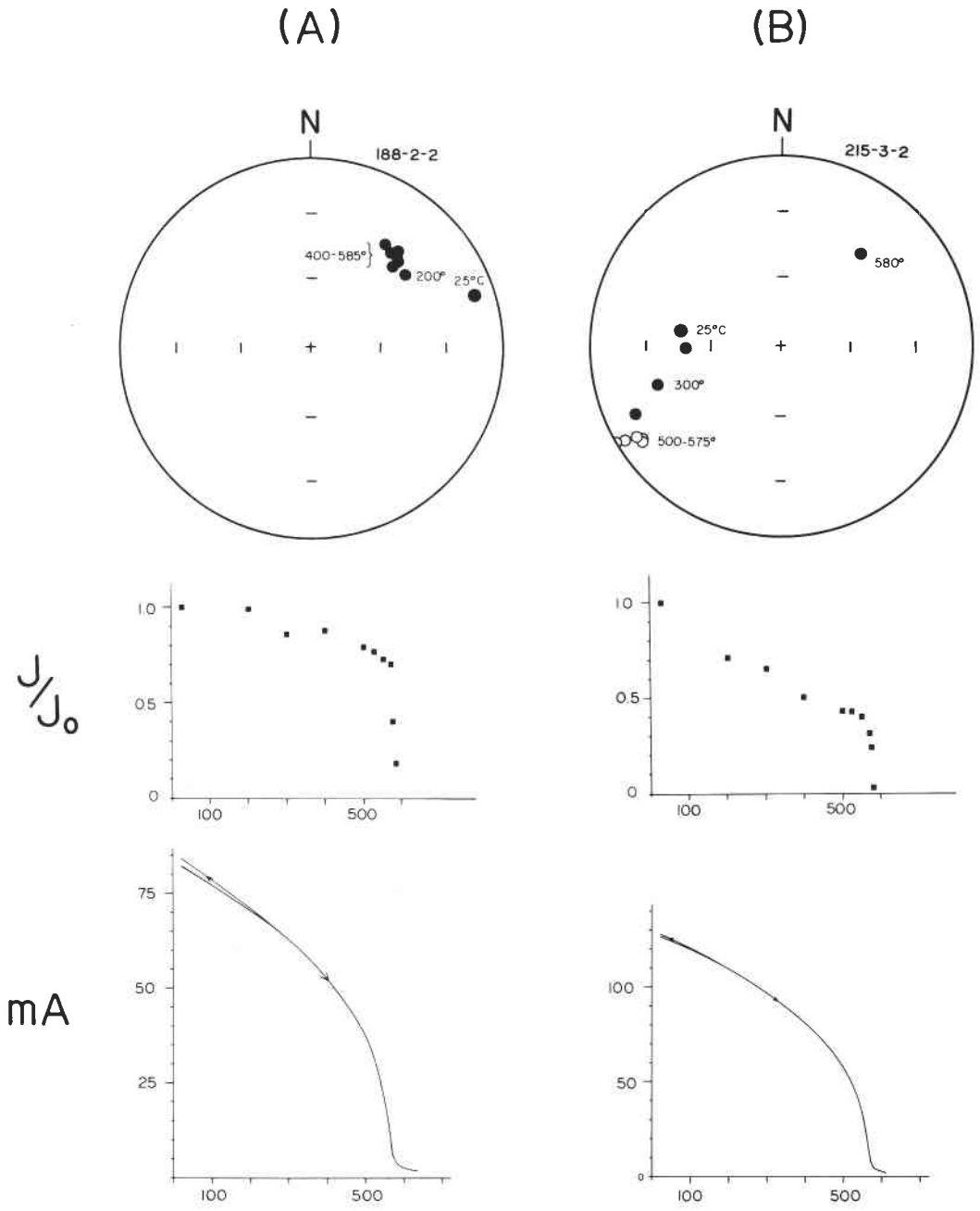


FIG. 6. The behaviour of (A) a normally magnetized and (B) a reversely magnetized Matachewan dyke sample after thermal demagnetization and thermomagnetic analysis. Top: Stereoplots of remanence directions after stepwise thermal treatment where open (closed) circles represent upward- (downward)-pointing magnetizations. Middle: Normalized intensity plots, where J_0 is the magnetization intensity of the NRM and J is the intensity after each thermal step. Bottom: Thermomagnetic curves showing the variation of saturation magnetization (proportional to the current in milliamps through a coil providing a null restoring force) against temperature; note the similarity of the heating and cooling curves, indicating that heating has had no apparent effect upon the magnetic minerals.

anorthosite host rock (foliation dipping 30°NW) on both sides of the dyke, from distances of 3–18 m from the dyke, gives a steep upward direction to the northwest (Fig. 10). The only anorthosite sample less than 3 m away (sample 181-12 at 2 cm) has a direction that is relatively close to that in the dyke (Fig. 10). Thus the data at site 181 form a positive baked-contact test, and the remanence in the anorthosite beyond a distance of 0.23 times the dyke width predates that in the dyke. Anorthosite at a distance of about 20 km from the ILCZ yields a remanence (the "B component" of Constanzo-Alvarez and Dunlop 1988) whose mean direction ($D = 115°$, $I = -68°$, $N = 4$, $k = 17$, $\alpha_{95} = 17°$) is not significantly different from the mean Kapuskasing dyke direction (Table 1). Hence our interpretation of the data at site 181 is that upon uplift through the Curie-point isotherm for

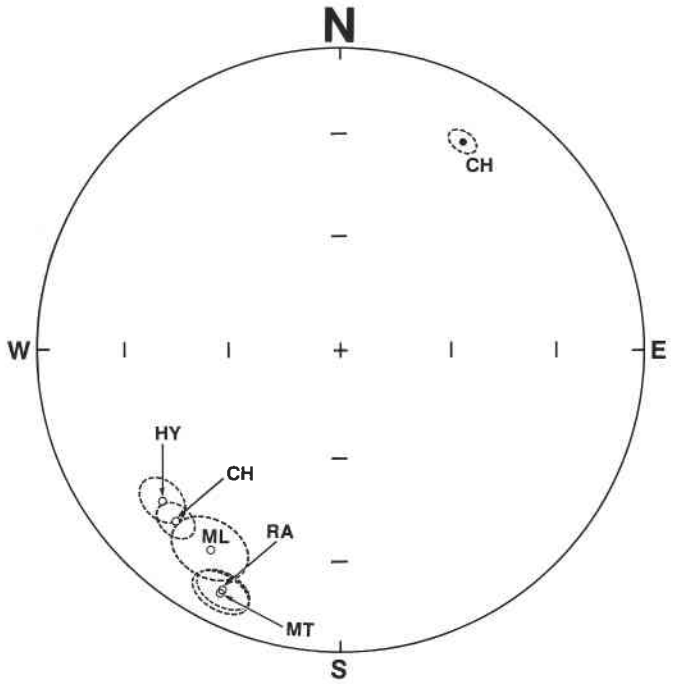


FIG. 7. Summary of mean paleomagnetic directions for the different areas where Matachewan dykes have been studied. HY, Highways 101 and 651 west of Chapleau; CH, Chapleau region; ML, Missinaibi Lake; RA, Ranger Lake (Halls and Shaw 1988); MT, Matachewan (Irving and Naldrett 1977; Pesonen 1973). Open (closed) circles represent upward- (downward)-pointing magnetization directions.

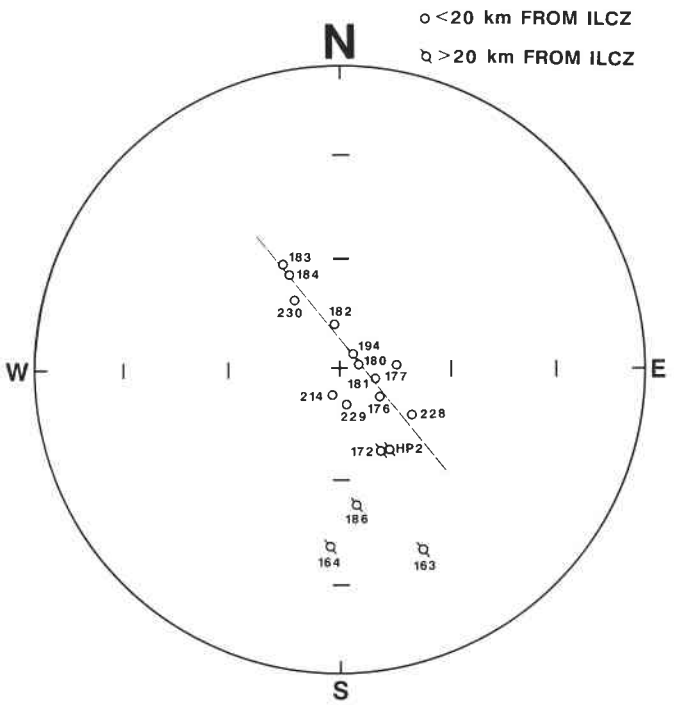


FIG. 8. Equal-area plot of site-mean remanence directions from east-northeast-trending Kapuskasing dykes. Site 184, in granulite host rocks, is also plotted for comparison. Data from dykes within 20 km of the ILCZ cluster about $D = 130°$, $I = -81°$ but also exhibit a pronounced streaking towards the northwest. Data from dykes more than 20 km from the ILCZ plot south of the main cluster.

magnetite (about 580°C) the anorthosite acquired a remanence and then was tilted to the northwest prior to the intrusion of the Kapuskasing dyke.

Along Missinaibi Lake, east-northeast-trending dykes are observed, but apart from one 40 m wide dyke near site KS4 (Fig. 2), the remainder are less than 5 m thick. Several of the Matachewan dykes are heavily overprinted by a remanence similar to that in Kapuskasing dykes (e.g., sites KS3, KS5, KS9 and KS11, Appendix), but only at site KS3 was the younger dyke exposed. Either a much greater concentration of Kapuskasing dykes exists along Lake Missinaibi than outcrop or aeromagnetic data suggest, or a major fault or fracture zone is present that provided access to fluids during the time of Kapuskasing dyke intrusion.

The present geographical limits of Kapuskasing dykes are approximately Lake Missinaibi to the northwest and the ILCZ to the southeast. South of the ILCZ, most east-northeast-trending dykes (1.14 Ga Abitibi; 2.14 Ga Preissac) yield a remanence with a steep positive inclination and westerly declination. Curiously, the KSZ contains two dyke swarms (Matachewan and Kapuskasing) that possess magnetic polarities opposite to those displayed by the majority of dykes of comparable trend and age that lie outside the zone!

Discussion and interpretation of results

Matachewan dykes

Magnetic-polarity distribution

A major discovery of this study is that there is an area of N-polarity dykes, approximately coincident with the KSZ, that is surrounded on all sides by Matachewan dykes having a dominant R signature. There are at least four possible explanations for this relationship. In order of preference they are as follows:

(1) The petrographic appearance of N dykes in the Chapleau area suggests that they crystallized at deeper crustal levels than those of dominantly R polarity outside the KSZ. The observed cross-cutting relations indicate that the N dykes are younger and presumably occupy a crustal level below and partly overlapping that of the R dykes. Dykes become more attenuated at increasing depths because within granulite host rocks of the KSZ both R and N dykes are absent, and in the adjoining amphibolite terrane only N dykes occur that are relatively thin (Appendix) compared with Matachewan dykes outside the KSZ. The change from predominantly R dykes outside the KSZ to N dykes within is therefore explained by uplift along the KSZ boundary faults and removal of the R dykes by erosion. Along Highway 101, both polarity boundaries also approximately coincide with mapped faults (Fig. 3), at least one of which is a late or post-Archean brittle-deformation feature (Moser 1988).

(2) The N-polarity dyke zone is continuous along the Matachewan swarm, but its full extent has not been realised because of inadequate sampling. The present data suggest that if it does continue it must be laterally offset along faults on either side of the KSZ. Based on the known distribution of R dykes along Lake Missinaibi and in the Hornepayne area, the only location for a swarm of N-polarity dykes north of the KSZ would be approximately east of the R dyke at site HP1 near Kapuskasing Lake (Fig. 2). A dextral offset of at least 10 km along an extension of the Wakusimi River fault (WRF in Fig. 2) could explain the anomaly. On the basis of reprocessed aeromagnetic data, West (1988) advocated increasing dextral offset of the Matachewan swarm towards the northeast across the KSZ. The dykes east of Hearst (H in Fig. 1) could thus be the offset equivalents of those in the Chapleau (CH) area, and the relatively dyke-free zone east of these dykes would thus continue across the KSZ. Therefore the absence of dykes in the

TABLE 1. Summary of paleomagnetic site-mean data for different regions

Area location	N	D (°)	I (°)	k	α_{95} (°)	Reference[a]
MATACHEWAN DYKE SWARM						
REVERSED-POLARITY DATA						
Missinaibi Lake (ML)[b]	6	213.7	−22.2	57	9.0	1
Highway 101 (HY)[c]	12	231.7	−22.2	53	6.0	1
All sites in KSZ region (CH)	18	225.7	−22.4	37	5.8	1
Ranger Lake (RA)	17	206.3	−12.5	35	6.1	2
Matachewan (MT)	11	206.0	−11.2	53	6.4	3, 4
NORMAL-POLARITY DATA[d]						
Zones parallel to and successively farther northwest of the ILCZ						
1. ILCZ to Lake Borden[e]	11	28.0	23.7	73	5.4	1
2. Lake Borden to Lake Kapuskasing	13	29.0	19.1	109	4.0	1
3. Lake Kapuskasing to Lake Missinaibi	13	33.8	24.2	51	5.8	1
All N sites in KSZ region (CH)	37	30.1	22.4	66	2.9	1
Zones parallel to dyke trend, listed from west to east						
4. Site E14 to site 219[f]	10	33.1	23.0	51	6.8	1
5. Site 220 to site 159[f]	11	24.0	21.2	142	3.8	1
6. Site E18 to site 227[f]	12	30.7	22.3	96	4.5	1
7. Site KS4 to site KS11[g]	4	40.2	21.6	63	11.6	1
KAPUSKASING DYKE SWARM						
Sites 172, 176–181, 194, 214, 228, 229, and HP2	10	130.3	−81.1	37	8.1	1
All sites	16	145.7	−80.0	11	11.9	1

NOTES: Matachewan directions have been recomputed for the location at latitude 47°N, longitude 84°W. N, number of sites; D and I, mean declination and inclination of the remanence direction.

[a]1, this study; 2, Halls and Shaw (1988); 3, Irving and Naldrett (1977); 4 Pesonen (1973).

[b]Site KS2 excluded, HP1 included.

[c]Sites 198 and 213 excluded.

[d]Site KS3 excluded.

[e]The boundaries of the zones are defined approximately by a northeast-trending line, parallel to the ILCZ, through the indicated lakes. The boundary through Kapuskasing Lake, between zones 2 and 3, is equivalent to the Wakusimi River fault that in part defines the northwest boundary of the KSZ. The boundary separating zones 1 and 2, through Borden and Nemogosenda lakes, may in part be a fault that locally separates granulite-grade rocks of the southernmost KSZ from lower grade rocks to the west.

[f]These zones exclude data from the Lake Missinaibi area.

[g]Although correlation is uncertain, these dykes may be equivalent to those of zone 4.

high-grade zone within would be because they were never emplaced there even at higher crustal levels. However, southwards across the ILCZ there is no obvious displacement in the dyke pattern, and yet there is a sharp mismatch in magnetic polarity. A sinistral offset along the ILCZ of at least 30 km is required, based on aeromagnetic data and the distribution of N and R dykes along Rocky Island Lake (Bates and Halls 1990). However, independent geological evidence for major transcurrent faulting along either of the two KSZ boundary faults has yet to be proposed.

(3) Matachewan dykes acquired an R magnetization first at shallow depths, where crustal host rocks were cooler, and then an N magnetization in their deeper roots as cooling migrated to deeper crustal levels upon uplift. The N dykes would thus represent the eroded roots of dykes that had R polarity higher up. The presence of amphibole rims around pyroxene could indicate that the N remanence was acquired upon cooling from amphibolite metamorphic conditions. However, remanence directions from host rocks in the Racine Lake area (Fig. 2) give a mean direction ($D = 31°$, $I = 53°$) distinct from that of N dykes in the same general area ($D = 31°$, $I = 22°$; zone 6, Table 1) and are interpreted as having been caused by uplift about 2.55 Ga ago before the dykes were injected (Constanzo-Alvarez and Dunlop 1988). Therefore, the lack of a common direction in dykes and host rocks suggests that the dykes did not become normally magnetized on being raised from crustal depths where amphibolite conditions prevailed.

(4) The N dykes represent localized igneous activity, centred on the KSZ, in which case transcurrent faulting is not required. This model might be acceptable if it could be demonstrated that the dykes were generated by an early thermal phase of KSZ activity. U–Pb dates on sphene (Percival *et al.* 1988) suggest that the deeper crust within the KSZ was at about 600°C during the time the Matachewan dykes were injected. However, the eastward decrease in ages towards the ILCZ, determined from a number of isotopic systems and minerals, suggests that the time when the sphene blocking temperature was reached was a consequence of a general cooling of the crust and was not caused by a thermal anomaly associated with the KSZ (Percival *et al.* 1988). Therefore, the lack of any thermal event accompanying KSZ development at about 2.5 Ga suggests that a spatially associated dyke swarm of this age is unlikely.

Estimates of crustal rotation along the Kapuskasing Structural Zone

One of our aims has been to estimate the amount of crustal rotation associated with deformation along the KSZ. However, for Matachewan dykes a clearly anomalous remanence direction

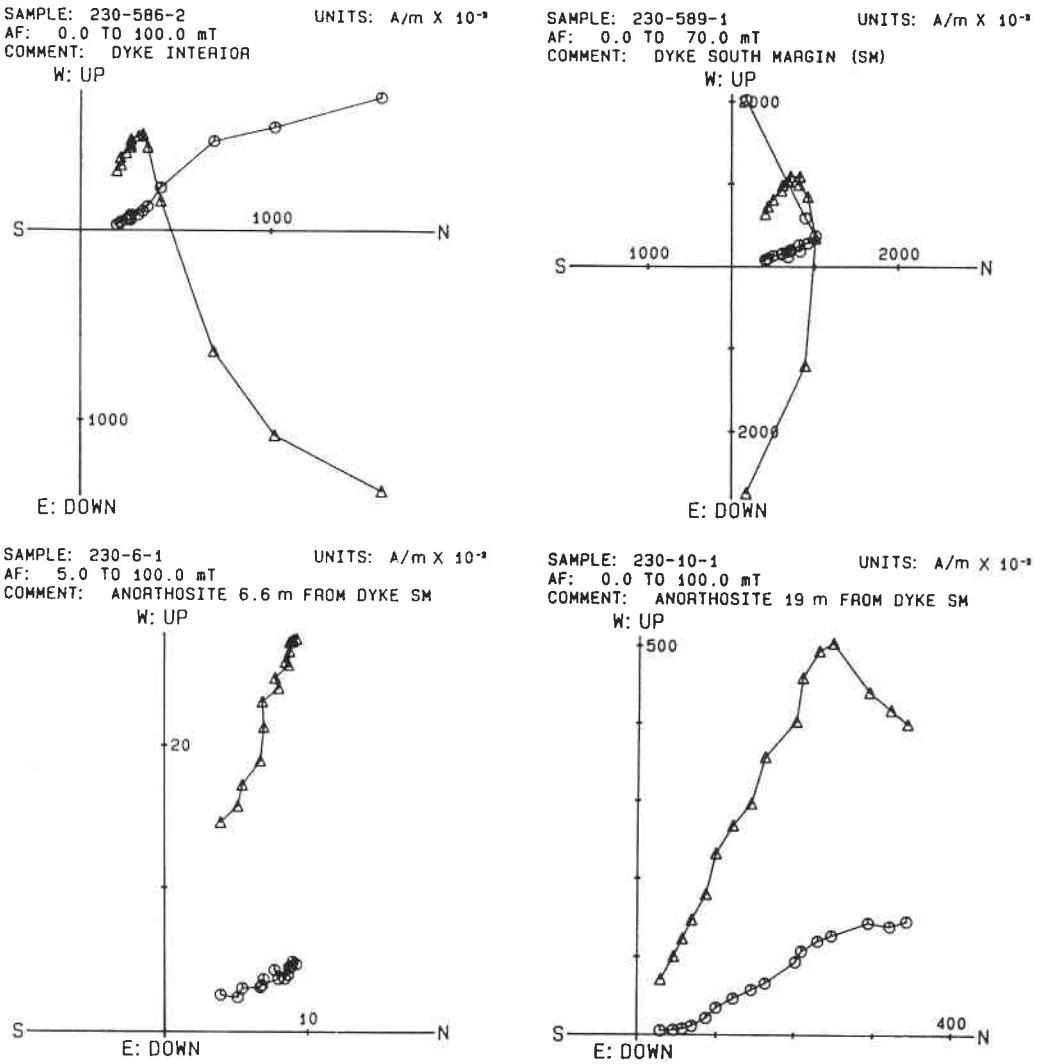


FIG. 9. Vector diagrams after AF demagnetization for samples from a 13 m thick Kapuskasing dyke at site 230 and its anorthosite host rock. Similar directions are obtained both in the dyke and in the anorthosite at distances of up to 1.5 times the dyke width. Symbols as defined in Fig. 5.

associated with the KSZ has not emerged (Fig. 7). The problem is compounded because reversal symmetry is not always a feature of the two polarities found in any one region (e.g., the CH data in Fig. 7), so in terms of crustal rotation the dominant R polarities outside the KSZ cannot strictly be compared with N data within.

If crustal tilting about north-northeast axes has occurred adjacent to the ILCZ on its hanging-wall side, the dykes should dip to the east and their remanence inclinations may become shallower, although the direction of the characteristic component is likely to be close to the rotation axis and thus not a sensitive indicator of crustal tilting. The dykes indeed have easterly dips from about site 214 eastwards, with only two exceptions (sites E18 and 226 in Fig. 3). To test for regional variations in paleomagnetic direction within the N population, we subdivided the data in two ways, firstly along zones parallel to the KSZ and at successively greater distances from the ILCZ (zones 1–3 in Table 1) and secondly along zones parallel to the trend of the dykes (zones 4–6, Table 1). In either scheme we see no evidence of anomalous directions on approaching the KSZ. At the 0.05 probability level, dykes around Chapleau are not significantly different from their N counterparts in the Lake Missinaibi – Highway 651 areas (see Table 1). The data means from zones 4–6 suggest that rather than being related to the KSZ, the directional differences reflect secular variation between dyke subsets within the N-polarity zone.

Kapuskasing dykes

Our paleomagnetic data suggest that the remanence in Kapuskasing dykes was acquired before and after crustal uplift and tilting adjacent to the ILCZ. Two episodes of dyke emplacement are proposed because the dyke at site 230 contains quartz and fails a contact test, whereas that at site 181 contains olivine and passes the test. The similarity in the paleofield direction at the time of their remanence acquisition might suggest an elapsed time of only several million years between the uplift and the later episode of dyking. An alternative interpretation is that the dyke at site 181 is approximately the same age as the 1.1 Ga Abitibi dyke of similar trend (Fig. 1) but opposite magnetic polarity. The presence of Middle Proterozoic igneous activity is suggested by east-northeast-trending dykes (sites 163, 164, 172, 186, and HP2) with shallower inclinations than found in other Kapuskasing dykes. They occur more than 30 km to the northwest of the ILCZ and may be related to the

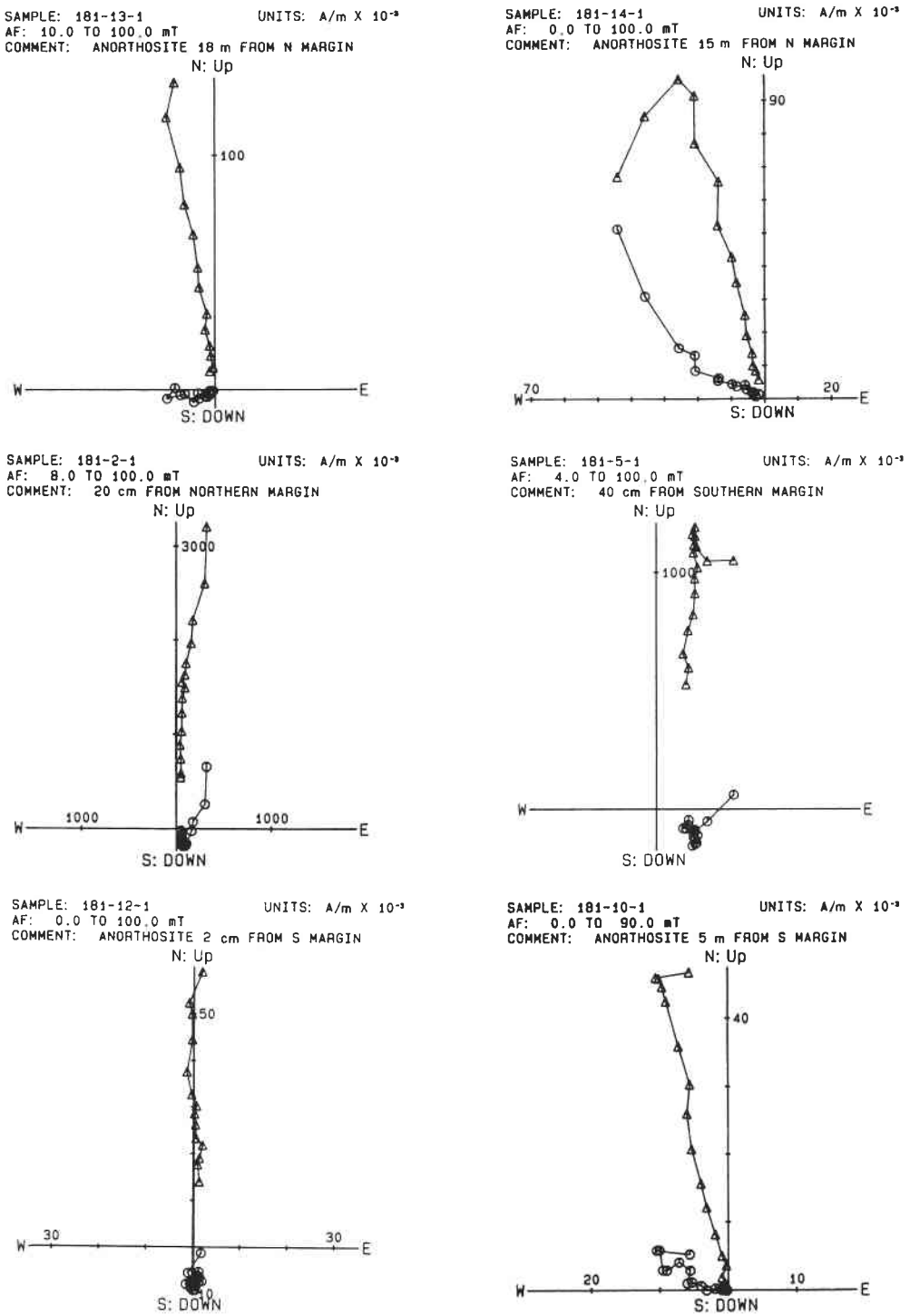


FIG. 10. A profile of selected vector diagrams after AF demagnetization taken from north to south (reading left to right down the diagram) across the 13 m thick Kapuskasing dyke at site 181 and adjacent anorthosite. The anorthosite sample 2 cm from the dyke has a remanence direction similar to that carried by the dyke. The remaining anorthosite samples at distances ranging between 0.23 and 1.5 times the dyke width have a consistent northwest-directed remanence, in contrast with the southeast declination found in the intervening dyke. Symbols as defined in Fig. 5.

1.0–1.1 Ga alkaline complex at Shenango River east of Lake Kapuskasing (Fig. 2) that has a similar paleomagnetic direction (V. Constanzo-Alvarez, personal communication, 1988).

Summary and conclusions

Within the KSZ, where Archean rocks are in upper amphibolite to granulite grades of metamorphism, both Matachewan and Kapuskasing dykes possess cloudy, tea-coloured plagioclase, a feature that may be diagnostic of dyke emplacement under conditions of elevated pressure and slow cooling that characterize deeper crustal levels. Significantly, cloudly feldspars are not found in Matachewan and Kapuskasing dykes intruding amphibolite- to greenschist-grade country rocks outside the KSZ. Two negative baked-contact tests on Kapuskasing dykes indicate uplift from depths where ambient temperatures were about 600°C. The collective petrological and paleomagnetic

data therefore suggest that within the KSZ both Matachewan and Kapuskasing dykes have been raised from crustal depths in excess of 10 km.

The presence of Kapuskasing dykes containing amphibole in the granulite-grade terrane of the KSZ may indicate that some crustal uplift had already occurred prior to their intrusion and therefore that the dykes were emplaced in metamorphic conditions that were somewhat less than granulite. It is possible by analogy with the Matachewan swarm that the quartz-bearing Kapuskasing dykes represent a deep crustal equivalent of the much wider and more widespread 2.14 Ga Preissac dykes that occur in lower grade terrane to the south of the KSZ and have a similar trend and age.

Our paleomagnetic data from the Matachewan swarm show a major anomaly within the KSZ: all the dykes in the Chapleau area have N magnetization, in sharp contrast with dykes along strike to the north and south, outside the KSZ, that have dominantly R polarity. Both the N and R magnetizations appear to be primary, and on the basis of cross-cutting relations from Lake Missinaibi the N dykes are younger. The boundaries between the N- and R-polarity domains appear to coincide with faults not only across the KSZ but along Highway 101 to the west. Our interpretation of the foregoing observations is that the N dykes have been relatively uplifted along faults and, compared with the R dykes, represent a deeper expression of the Matachewan swarm. *If* the N magnetization is primary, the swarm is constructed from the top down by two partly overlapping populations of dykes having a blade-like form that occupy discrete levels within the crust, the younger, deeper one pinching out at a depth above that corresponding to granulite-grade conditions. The absence of anomalous paleomagnetic directions in the Matachewan dykes on approaching the ILCZ suggests that crustal tilting after dyke emplacement was minimal, perhaps less than 10° at distances more than 4 km from the fault zone, or that crustal tilting was greater but the rotation axis was closely aligned with the remanent-magnetization direction.

In conclusion, petrographic and paleomagnetic data from two Early Proterozoic dyke swarms suggest that major crustal uplift along the KSZ occurred after the dykes were emplaced. We have discovered domains of contrasting magnetic polarity within the Matachewan swarm that are related to vertical fault displacements. Their further definition may be of considerable importance in identifying major faults in poorly known gneissic areas of the Superior Province where lithological contrasts are small.

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Appendix

TABLE A1. Summary of all structural and magnetic data

Site	Azimuth of dip (°)	Dip of dyke (°)	Width of dyke (m)	N	N_S	P	D (°)	I (°)	α_{95} (°)
			Matachewan–Hearst dykes						
KS1	85	70	11	7	7	0	31	18	4
KS2	80	85+	43	11	2	1	247	−8	31
KS3[a]	70	90	44	8	1	—	28	37	—
KS4[a]	80	85+	27	17	7	½	221	−15	12
KS4[a]	240	85	4	16	8	0	28	22	7
KS6	60	90	20	9	5	—	217	−24	9
KS7	80	90	24	10	1, 5	½	200	−27	25
KS8[a]	80	85+	21	8	5	1	220	−19	7
KS9	80	85+	21	11	3	0	46	24	11
KS10	95	65	14	10	9	—	37	29	6
KS11[a]	265	65	23	16	6	1	223	−31	12
KS11[a]	265	—	0.02	6	6	0	51	19	9
HP1	75	90	>18	8	5, 3	—	202	−28	15
HP17	60	80	>1	7	7	0	22	34	4
HP18	60	90	10	8	8	0	28	33	5
E18	95	80	9	3	3	0	32	20	7
E19	105	80	>20	5	5	1	37	15	5
159	80	90	~3	5	5	0	13	25	11
160	80	75	8.5	5	5	1	20	27	9
161	80	80	4.2	7	7	0	34	19	4
168	90	—	20	7	7	0	28	21	8
169	85	—	21	12	10	—	35	17	6
170	85	90	16	12	11	0	34	24	5
173	75	—	~14	5	3	½	39	26	7
174	85	80	7.8	7	6	—	35	23	11
185	30	90	37	7	4	0	222	−25	16

TABLE A1 (*continued*)

Site	Azimuth of dip (°)	Dip of dyke (°)	Width of dyke (m)	N	N_S	P	D (°)	I (°)	α_{95} (°)
E14	55	85	26	7	2, 3	0	26	19	12
186[a]	65	90	30	9	5	0	33	21	14
188	260	85+	21	5	4	1	38	27	8
189	250	80	23	6	5	0	33	21	7
191	75	80	6	9	7	0	22	34	7
195	265	85+	19	8	5	0	215	−35	6
196	265	70	31	7	2	0	231	−20	10
197	270	85+	21	9	5	1	244	−28	6
198	265	85	25	3	1, 1	0	220	−33	—
205[a]	260	75	24	12	7	0	236	−33	8
206	270	85	8.8	7	4	½	39	44	8
207	95	80	2.5	3	3	1	245	−19	19
208	80	85+	~20	8	2, 2	1	240	−16	7
209[a]	95	90	17	8	6	1	221	−24	14
210	270	75	>21	4	3	1	232	−21	17
212	95	—	>13	6	3	1	229	−25	22
213	—	—	>10	3	1, 1	1	217	−21	—
214[a]	65	80	29	12	10	0	32	31	6
215	85	—	~30	6	5	0	228	−20	4
216	90	80	19	6	3	0	237	−9	15
217	—	—	>15	3	3	0	36	19	6
218	85	80	16	7	4	1	33	17	10
219	90	85	>22	4	3	0	40	11	12
220	80	85+	4	7	7	0	28	20	6
221	90	—	12	7	6	1	26	24	11
222	85	80	20	5	5	0	20	19	4
223	90	85+	15	4	2	0	25	22	3
224	80	75	~18	5	5	0	20	19	9
226	90	85+	17	6	6	0	31	27	3
227	80	—	>1.7	5	5	1	20	30	5
231	85	90	27	7	4	½	24	17	11
232[a]	70	—	15	6	4	1	28	25	5
233	60	65	14	8	6	½	31	31	5
187	80	85	~20	1	0	0			
190	70	85	7	4	1	—	N polarity		
193	80	85	~15	3	1	—	N polarity		
199[b]	60	90	18	8	1	1	N polarity		
200[b]	280	80	30	3	0	0			
201[b]	280	80	23	0	0	1			
202	55	45	12	3	0	—			
203	80	90	12	2	1	1	R polarity		
204[b]	65	80	19	7	1	½	N polarity		
211	Variable		3	3	1	0	R polarity		
225	85	85	14	6	0	1			
			Kapuskasing dykes						
HP2	340	85	34	4	4	0	149	−65	9
163	150	75	15	7	6	0	155	−35	5
164	330	85	>6	6	6	0	184	−43	29
172	160	—	>21	11	11	0	153	−66	5
176	180	65	~5	7	3	0	125	−77	24
177	170	90	12	7	4	0	87	−75	17
180	330	85+	13	7	7	0	79	−85	3
181	170	80	13	9	9	0	105	−80	4
181[c]	330	30	—	5	4	—	285	−79	6
182	160	70	12	6	6	1	356	−72	8
183	155	65	13	8	8	0	334	−57	7
184[d]	—	—	—	10	10	—	335	−60	13
186[a]	150	90	10[e]	12	9	0	173	−54	9
194	175	80	12	7	6	½	47	−85	4
214	355	80	8[e]	7	7	0	196	−83	5
228	165?	—	>9	5	4	0	122	−67	5
229	160	70	>4	6	6	0	167	−81	10
230	165	75	13	8	8	0	338	−60	6
230[c]	330	40	—	9	5	—	329	−68	10

TABLE A1 (*concluded*)

Site	Azimuth of dip (°)	Dip of dyke (°)	Width of dyke (m)	N	N_S	P	D (°)	I (°)	α_{95} (°)
			Unclassified dykes						
KS3[a]	130	85	17	7	7	0	87	−71	8
KS8[a]	130	90	0.7	2	2	0	335	43	9
162	310	80	2	9	8	0	153	00	19
165	265	60	7.5	9	9	0	112	−49	9
167[a]	165	—	3.5	12	11	0	357	55	16
175	175	90	>9	8	8	0	296	−12	16
			Magnetic overprints						
KS5[f]	70	85	25	5	4	0	97	−63	6
KS9[f]	80	85+	21	2	2	0	154	−48	14
KS11[f]	265	65	23	3	3	0	75	−74	18
171[f]	250	80	30	5	4	0	129	−44	21

NOTES: N, total number of samples measured at a site; N_S, number of samples giving stable end points followed, where indicated, by the number of additional samples yielding remagnetization circles; P, degree of feldspar phenocryst development where 0 = phenocrysts absent or occasionally seen, ½ = scattered phenocrysts, and 1 = phenocrysts abundantly developed either locally or throughout the dyke; D and I, site declination and inclination of the characteristic remanent magnetizatiion; α_{95}, radius of the 95% confidence cone about the mean direction.

[a]The Matachewan dyke is cut by another dyke.

[b]The dyke is heavily altered with epidote and (or) chlorite and (or) quartz veining.

[c]Anorthosite.

[d]Mafic gneiss.

[e]The dyke width is the cumulate value of more than one dyke.

[f]The Matachewan dyke shows a pronounced "Kapuskasing dyke" overprint, although no younger dyke is seen.